DB-Curator: On-demand scientific database curation via deep-research multi-agent literature mining and verification

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Abstract

Scientific databases represent a core foundation of scientific knowledge, yet their development remains time consuming, difficult to scale, and hard to reproduce across specialized topics. Existing automated assistants can answer questions or summarize papers, but they do not reliably construct topic-specific databases with entry-level verification and auditable provenance, especially when the relevant literature set must be built from the user request rather than supplied by human curators. Here we present DB-Curator, a multi-agent system that constructs topic-specific databases from open-ended scientific requests. DB-Curator converts the user request into a machine-actionable rubric, builds a dynamic literature set, discovers candidate entities, and verifies each candidate through a Researcher–Critique loop. We evaluated DB-Curator in two complementary settings. First, in fixed-cohort release audits, DB-Curator produced 246/250 release-ready rows and recovered 68 literature-supported compounds that were absent from the benchmark but judged valid by human experts. These entries were not only plausible names: all 2166 cited primary-paper references in this audit resolved to real papers and contained the quoted evidence span or equivalent local context. The zero-shot direct-generation baseline produced many acceptable names, but only 11/257 claimed references provided usable paper evidence. Second, across five benchmark-scale verification tasks spanning 1,887 candidate rows, DB-Curator assigned Confirmed labels to 977 entries, compared with 375 entries assigned Confirmed by the zero-shot benchmark-labeling baseline. DB-Curator

also maintained quote- or context-supported provenance for 7,430 of 7,453 cited primary-paper references, whereas the strict baseline citation audit found content support for only 6/2,291 claimed references. These results show that DB-Curator improves both evidence-grounded discovery of new database entries and large-scale confirmation of existing candidate sets, supporting predicate-grounded, critique-driven database construction as a general framework for scalable scientific curation.

Main

Scientific databases make experimental findings searchable, comparable, and reusable across studies. Conventionally, expert curators define the database scope, search the literature, identify candidate entities, normalize names and identifiers, read papers for inclusion evidence, and record provenance for later review. This process is powerful but slow, tedious, and difficult to repeat when the target topic is narrow and the evidence is scattered across many papers. In metabolomics and exposomics, resources such as HMDB and the Blood Exposome Database demonstrate the scientific value of expert curation, while recent work on disinfection-by-product (DBP) resources underscores how labor intensive topic-specific database construction remains.[1-3] More broadly, resources such as T3DB, PubChem, and DrugBank show that useful databases require coverage, stable identifiers, explicit inclusion criteria, and maintainable provenance.[4-6] Scientific database construction is therefore a natural target for automation, but only if automated systems can recover candidates broadly and justify every accepted entry with auditable evidence.

Several developments make automated curation more realistic. Open literature infrastructure has expanded programmatic access to full text and metadata through Europe PMC, S2ORC, OpenAlex, and related resources. Document-processing systems such as PubTator and GROBID now support structured parsing of biomedical literature.[7-12] Automated extraction pipelines have also improved substantially. Chemical text-mining tools and benchmarks such as ChemDataExtractor, CHEMDNER, and tmChem established strong baselines for named-entity recognition and normalization, and domain-adapted models such as BioBERT, SciBERT, and HunFlair2 have strengthened scientific language understanding.[13-18]

These advances do not by themselves solve release-quality database curation. A chemical name in a paper is not automatically a valid database entry. For example, a candidate may be a reagent rather than a measured analyte, a precursor rather than a disinfection byproduct, a compound reported in another biological matrix rather than blood, or a class-level mention rather than a specific confirmed entity. Inclusion therefore depends on topic-specific predicates such as identity, experimental context, sample matrix, analytical confirmation, formation route, organism, exposure source, or exclusion criteria. These predicates may be distributed across titles, abstracts, methods, tables, and result sections. Errors at this stage can propagate into curated resources, where they become costly to detect and correct.[19-21] Large language models (LLMs) broaden what is possible

in scientific reading and information extraction,[22-24] but direct single-pass use remains risky for database construction. Plausible but unsupported statements can become false entries unless provenance and acceptance rules are enforced explicitly.[25-27]

This creates a methodological challenge that is different from standard retrieval-augmented generation. A conventional RAG system typically retrieves a context set and then generates an answer from that context. Scientific database construction instead requires repeated testing of a candidate against a user-defined acceptance contract. Missing evidence, exclusion-triggering evidence, and off-scope context must change the decision for that entry rather than merely change the wording of a generated answer. The core AI problem is therefore not only retrieval quality. It is how to convert natural language intent into explicit predicates that can be checked, revisited, and left evidence-limited when support remains insufficient.

Single-agent and multi-agent LLM systems provide useful building blocks for this problem. Scientific claim verification treats correctness as evidence retrieval plus rationale selection over the literature.[28-31] Agentic systems and deep-research workflows show how language models can interleave retrieval, reasoning, tool use, and self-correction over multiple steps.[32,33,39,40] Modular tool-use and skill-composition systems such as MRKL, Toolformer, and Voyager further frame language models as controllers over external capabilities and reusable action routines.[41-43] Prior scientific systems such as OpenChemIE, ProtAgents, MAIN-RAG, and tool-augmented chemistry assistants are important precursors, but they primarily emphasize extraction, question answering, or discovery support rather than evidence-constrained database construction from a literature set that is itself built during the run.[34-38]

Here we present DB-Curator, an agentic system designed specifically for scientific database construction. Its distinctive feature is the coupling of dynamic, user-conditioned literature construction with predicate-state verification. The initial user request determines broad searches, inspected search results seed new queries, candidate identities trigger entity-specific retrieval, and unresolved rubric predicates determine follow-up searches. The clarified user intent is decomposed into rubric items, and the agents iteratively update the state of each item from cited evidence until the candidate is accepted, excluded by rubric criteria, or left evidence-limited. Each output is a structured database entry grounded in supporting spans. Relative to prior agentic scientific assistants, DB-Curator is designed around entry-level acceptance decisions rather than free-form answers, generic retrieval-augmented answers, or document-level extraction alone.

Results

DB-Curator operationalizes curation as predicate-state verification

DB-Curator is designed to resolve the central tension in automated curation: broad discovery is needed to find candidate compounds, but release-quality database entries require strict evidence fidelity. The system addresses this

tension by treating each candidate compound as an evidence contract rather than as a generation target. Figure 1 summarizes the four operational stages: intent refinement into a topic rubric, dynamic literature construction and entity discovery, candidate pre-ranking, and entry-wise Researcher–Critique verification. The key methodological objects are the evolving literature set and the predicate state of each candidate. Required predicates must be supported by cited paper context, exclusion criteria must not be triggered, and unresolved predicates drive targeted follow-up retrieval or remain evidence-limited.

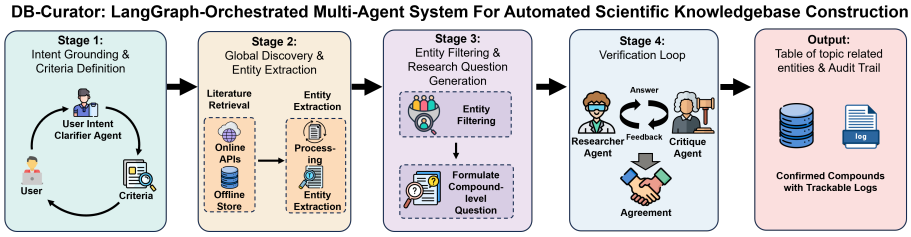


Figure 1 | System architecture and modular workflow. The DB-Curator framework orchestrates LLM agents and domain-specific tools via LangGraph.[44] It transforms unstructured research objectives into structured databases. The workflow starts with Stage 1, human-in-the-loop intent refinement, where a clarification agent iteratively formalizes user intent into explicit scientific criteria. Stage 2 performs agentic literature mining rather than extraction from a fixed corpus: broad queries, inspected search results, seed entities, and later unresolved rubric predicates all update the paper set used for curation. Retrieved papers are parsed with an OCR-aware extraction stack, using OpenDataLoader[48] as the primary PDF conversion path with Docling fallback[47] when needed, and are then processed with HunFlair2 followed by entity normalization. Stage 3 performs lightweight candidate pre-ranking before downstream review. Candidate mention snippets are scored against topic-related query text to identify semantically relevant candidates. Only the top-ranked subset is forwarded to LLM filtering and downstream verification. Stage 4 runs a critique-driven verification loop between a Researcher agent and a Critique agent. The Researcher enriches chemicals using PubChem, RDKit, and name-to-structure resolution when available and gathers evidence with adaptive query refinement and retrieval-augmented reading. The Critique agent checks claim–evidence alignment at the rubric-item level and requests targeted follow-up retrieval until the required predicates are either supported or exhausted.

Fixed-cohort full-pipeline comparison

We first asked whether the full DB-Curator workflow produces release-ready rows more reliably than direct generation from the same topic definition. This comparison is important because a strong LLM can often name plausible compounds, but database release also requires identity grounding and paper-level provenance. We compared DB-Curator against a zero-shot direct-generation baseline across all five topic-level human audits. DB-Curator used Gemini 2.5

Flash as the Researcher model and GPT-5-mini as the Critique model. The baseline used Gemini 2.5 Flash alone, without retrieval tools, compound grounding, or critique-loop verification.

Because curated benchmarks are incomplete for open-ended database construction, Figure 2 separates evidence grounding, release readiness, and curation effort. A benchmark-overlap row is a generated candidate whose normalized identity matches a benchmark entry. A release-accepted row is either a benchmark-overlap row or a non-overlap row judged true by manual inspection. Quote/context support means that the cited paper contains either the extracted evidence span or local context with the same support meaning. Figure 2A reports citation existence and quote/context support before the name-level analysis because evidence grounding is the release bottleneck. After excluding compound-database lookup records, 2166/2166 cited primary-paper references from DB-Curator resolved to real papers and 2166/2166 contained the quoted span or equivalent local context. By contrast, the zero-shot direct-generation baseline produced many plausible rows but only 11/257 claimed references resolved to papers with usable content support.

Figure 2B shows why benchmark overlap alone underestimates useful curation and why human review should focus on the non-overlap remainder. Across the five fixed cohorts, DB-Curator produced 246/250 release-accepted rows, comprising 178 benchmark-overlap rows and 68 non-overlap rows judged true by human inspection. These 68 rows are the important out-of-benchmark discoveries: they were not recovered by benchmark matching, but the compound identity and supporting evidence were accepted by human experts. The four remaining DB-Curator rows were non-overlap errors, including false compound assignments or out-of-scope precursors. Figure 2C shows that this reliability came from a selective candidate funnel: depending on the topic, DB-Curator evaluated 52–124 compounds to obtain each final 50-row accepted cohort. Figure 2D then evaluates the same release requirement at the entry level. All DB-Curator rows had real-paper support, whereas only a small fraction of zero-shot baseline rows had any resolvable reference trail. Together, these panels show that DB-Curator is not merely reproducing familiar benchmark entries. It recovers known compounds, surfaces literature-buried valid compounds outside the benchmark, and attaches auditable provenance to both.

The zero-shot Gemini 2.5 Flash baseline showed why name-level success is not enough for database construction. Under the same overlap-first and manual-inspection rule, 249/250 generated names were release-accepted, including 170 benchmark-overlap rows and 79 non-overlap rows judged true. However, the citation trail was weak: the baseline claimed 257 references, but only 11/257 could be resolved as real papers and at most 11/257 contained candidate-relevant evidence under the same quote/context-support criterion. Its mean existing-reference depth was only 0–0.10 references per row depending on topic. Thus the main difference was not whether a model could mention familiar compounds, but whether it could construct auditable, topic-specific, literature-grounded database

entries.

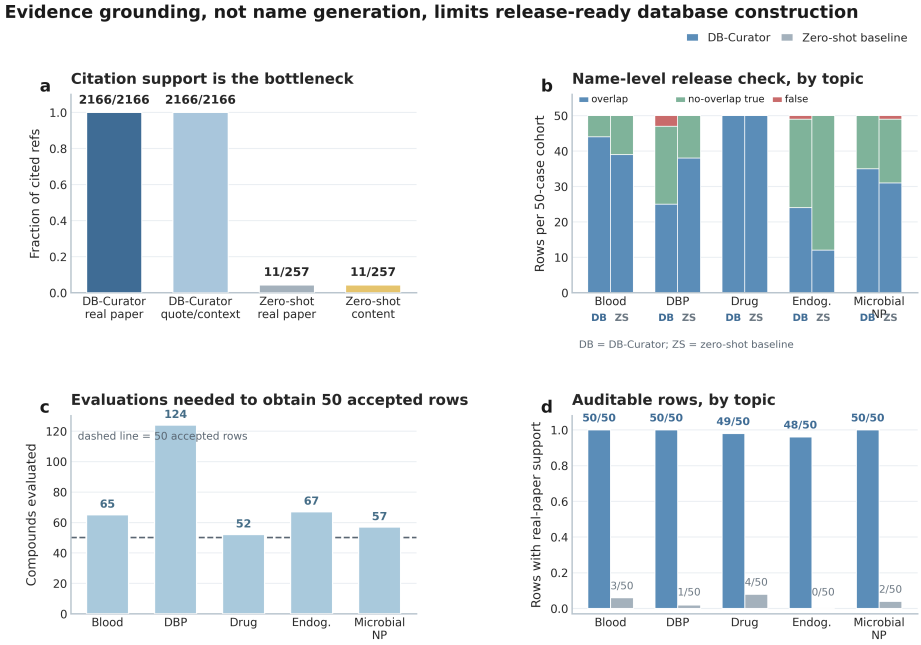


Figure 2 | Evidence grounding, rather than name generation, limits release-ready database construction.

A, Reference audit at the cited-paper level, measuring both paper existence and whether the quoted evidence span or equivalent local context was found in the paper. B, Topic-wise name-level release check across the five 50-row cohorts: within each topic, the left stacked bar is DB-Curator and the right stacked bar is the zero-shot baseline, partitioned into benchmark-overlap, no-overlap true, and false rows. C, Number of per-compound evaluations required to obtain 50 accepted DB-Curator rows in each topic. D, Entry-level fraction of rows with any real-paper support.

Verification Loop Benchmarking and Analysis

We next evaluated benchmark-scale verification across five source domains: DBPs from a systematic DBP curation resource,[3] blood exposome chemicals from the Blood Exposome Database,[2] drugs from DrugBank,[6] human endogenous metabolites from HMDB,[1] and microbial natural products from the Natural Products Atlas.[52] The DBP benchmark contained 687 candidate rows. For each of the other four domains, 300 cases were randomly chosen from the corresponding source database. In Figure 3B, only Confirmed rows are counted as database-ready positive entries. Suspected, Explicit Negative Evidence, and Insufficient Direct Evidence are grouped as not confirmed for this positive-entry analysis because they do not provide direct positive support for database inclusion.

Figure 3A shows the final DB-Curator outcome composition, separating Confirmed rows from entries that remained below the direct positive evidence standard. Figure 3B shows that the final DB-Curator labels are strongly domain-dependent. DB-Curator assigned Confirmed labels to 321/687 DBPs, 86/300 blood exposome chemicals, 210/300 drugs, 123/300 endogenous metabolites, and 237/300 microbial natural products. The zero-shot Gemini 2.5 Flash benchmark-labeling baseline assigned Confirmed labels to fewer cases under the same topic definitions: 137/687 DBPs, 5/300 blood exposome chemicals, 120/300 drugs, 69/300 endogenous metabolites, and 44/300 microbial natural products. Figure 3C then asks whether the cited evidence can support these decisions. It reports content-supported reference counts rather than another near-binary fraction. DB-Curator cited 7,453 paper references in this benchmark-scale analysis. Of these, 7,433/7,453 resolved to real papers and 7,430/7,453 contained the quoted evidence span or equivalent local context. This audit resolves cited items to papers and then checks available or recovered text for the quoted span or equivalent local context. Compound-database identity pages are not counted as primary papers. The matched zero-shot citation-support baseline produced a different failure mode from the fixed-cohort direct-generation baseline. In a permissive metadata-resolution audit, many claimed identifiers resolved to some bibliographic record, but title and context checks showed that this was not sufficient evidence support. Under the stricter benchmark-labeling citation audit used in Figure 3C, the zero-shot baseline produced 2,291 claimed references. Only 23/2,291 resolved to title- and identifier-compatible papers, and only 6/2,291 had candidate- and evidence-context support. This shows that the benchmark-labeling task is not just name recognition or citation plausibility. Evidence retrieval and quote-grounded verification are needed to apply topic-specific database criteria.

Figure 3D uses human verification for a different question. It audits the rows that DB-Curator resolved as Confirmed or Explicit Negative Evidence and asks whether the decision is aligned with the user-defined topic scope. This matters because a benchmark entry is not automatically inside the narrower user definition used for a particular run. For example, a legacy benchmark may include compounds under broader database conventions, whereas DB-Curator is asked to apply the explicit topic definition and rubric supplied for this study. Across 1091 audited Confirmed or Explicit Negative Evidence rows, 1085 were judged real, topic-aligned outcomes. Thus, negative labels on benchmark compounds are not necessarily system mistakes; they often indicate that the supplied compound did not satisfy the stricter user topic definition. The full topic definitions and rubric criteria are provided in the Supplementary Information.

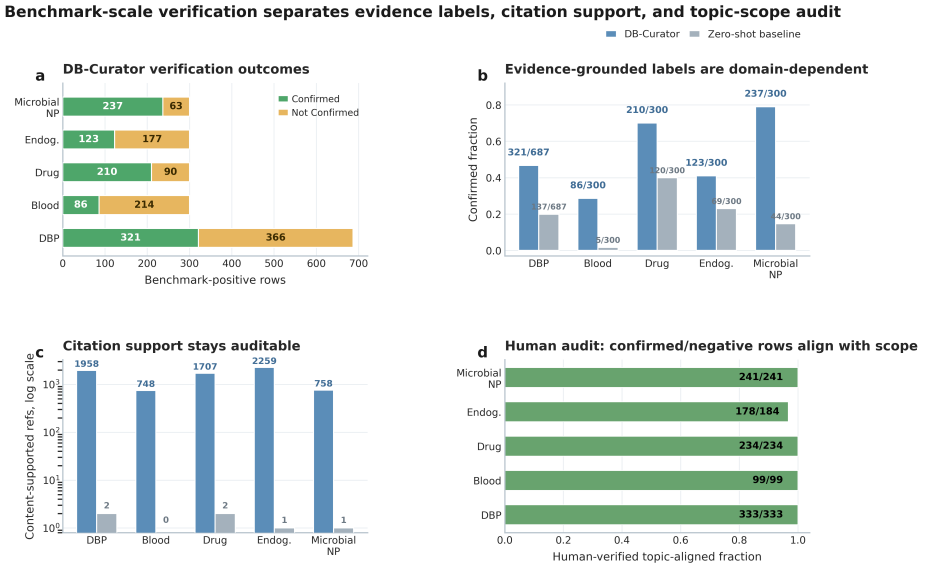


Figure 3 | Benchmark-scale verification, zero-shot baseline, and evidence grounding across five chemical domains.

A, DB-Curator benchmark-scale outcome counts, separating Confirmed rows from rows that remained below the direct positive evidence standard. B, Confirmed fractions assigned by DB-Curator and by the zero-shot baseline under the same topic definitions. C, Topic-wise content-supported reference counts for DB-Curator and the matched zero-shot citation-support baseline, shown on a logarithmic scale. D, Human analysis of rows resolved as Confirmed or Explicit Negative Evidence. Counts report how many audited rows were real, topic-aligned outcomes under the user definition.

Rubric Discrimination and Agent Routing

We then asked which rubric gates separate accepted compounds from rows that remain below the direct-evidence threshold. For each topic, we computed overall criterion pass rates and the number of criteria passed by each row. Figure 4 expands the rubric analysis across all five topics rather than collapsing the criteria into one global score. DBP confirmation is driven by direct formation, precursor identification, and drinking-water relevance. Blood exposome confirmation is driven by evidence for human blood or systemic circulation. Drug confirmation is controlled by drug-entity and formulation-exclusion criteria. Endogenous-metabolite confirmation is driven by human production, function, and exclusion of solely exogenous or microbiota-only explanations. Microbial-natural-product confirmation depends on analytical evidence, unambiguous structural assignment, and microbial origin. The left column reports how often each rubric gate is passed across the evaluated rows, with overlaid green and red markers showing the corresponding pass rates in Confirmed and non-Confirmed rows. The right column summarizes whether rows accumulate enough solved criteria

to support database-ready decisions. Together these panels make the predicate logic auditable. DB-Curator is not applying a single global confidence score; it is routing each topic through the gates that define database readiness for that scientific context. The result is an interpretable failure profile: unresolved DBP rows are often missing direct formation evidence, blood exposome rows are often limited by specimen-context evidence, and microbial-natural-product rows depend strongly on structural and origin support.

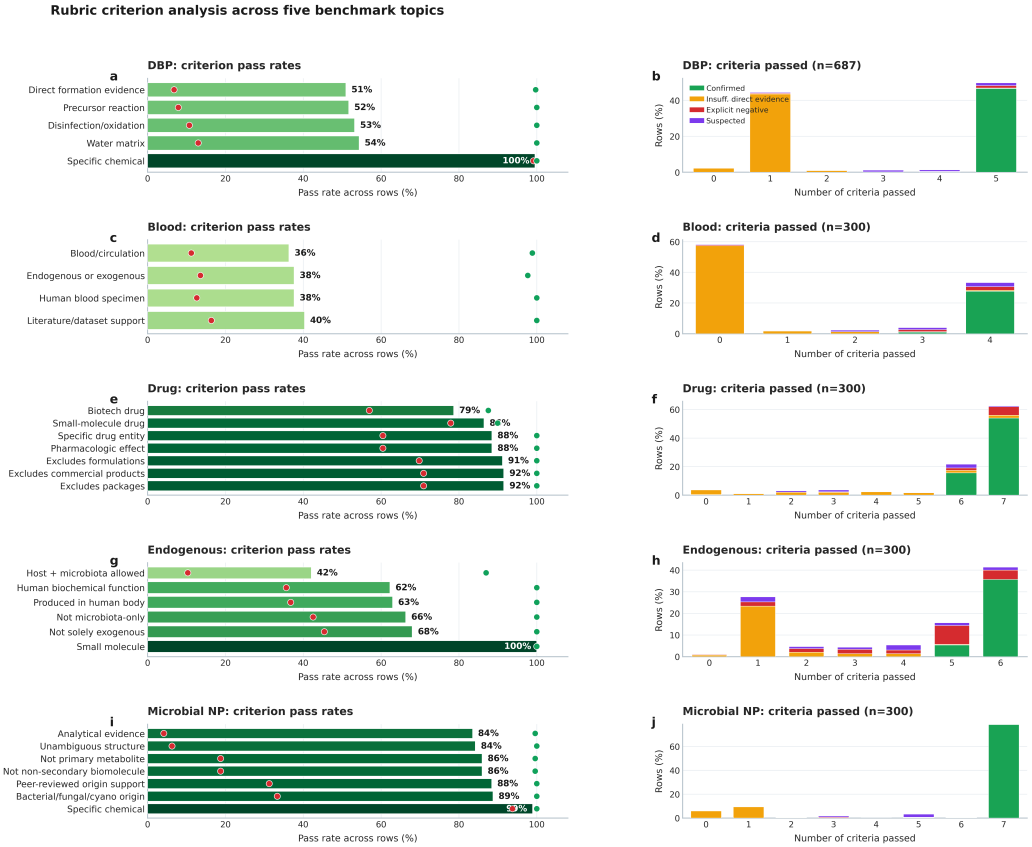


Figure 4 | Topic-specific predicate gates discriminate accepted entries from evidence-limited outcomes.

A–J, Rubric diagnostics across the five benchmark topics. For each topic, the left panel reports overall pass rates for the topic-specific rubric gates, with green and red markers indicating pass rates among Confirmed and non-Confirmed rows, respectively. The right panel shows the distribution of how many criteria were passed by rows assigned to each final label. The panels show that database-ready outcomes are separated by topic-specific predicate gates and structured failure patterns rather than a single global score.

Iterative Retrieval and Critique-Agent Utility

Finally, we isolated the role of retrieval mode and iterative critique. The comparisons separate direct generation, topic-definition labeling, researcher-only pre-critique behavior, the full Researcher–Critique loop, and retrieval-source breadth. Direct zero-shot generation produced plausible names but only 11/257 resolvable references. Zero-shot benchmark labeling assigned Confirmed labels to 375/1,887 candidates but yielded only 6/2,291 content-supported references under the same strict citation-support audit used for Figure 3C. In contrast, the full Researcher–Critique loop assigned Confirmed labels to 977/1,887 candidates with 7,430/7,453 quote/context-supported references. Figure 5A treats the first researcher claim as a researcher-only pre-critique state and compares it with the final critique-loop decision. Across 1,887 benchmark rows, the first researcher state assigned Confirmed labels to 933 rows, whereas the final loop assigned Confirmed labels to 977 rows. The loop rescued 49 rows to Confirmed and tightened 5 initially Confirmed rows away from Confirmed. Figure 5B evaluates these 54 confirmation-changing revisions against human inspection and shows both topic-wise counts and the pooled 51/54 right-label result. The 49 rescued-to-Confirmed cases were all judged right by human inspection. Among the five tightened-away cases, two became acceptable concrete negative labels, whereas Harmaline, 3-Hexanone, and 6-Succinoaminopurine were over-tightened to Suspected despite human inspection supporting a topic-acceptable final label.

Figure 5C shows that retrieval mode also changes the outcome. In a 50-case mode ablation, hybrid retrieval confirmed 16/50 cases compared with 12/50 for offline-only retrieval, indicating that web and local evidence sources are complementary rather than redundant. Hybrid retrieval required more compute, with median runtime increasing from 109 s to 555 s and median token use increasing from 51,024 to 77,343 tokens. Figure 5D shows why the additional compute can matter. Changed-claim rows required more researcher calls than stable rows (5.28 versus 2.84 on average), more critique calls (2.00 versus 1.27), and more supporting sources (6.40 versus 4.23), all shown directly in Figure 5D. These dynamics support the intended role of the Critique agent. As evidence accumulates, the system can revise a provisional claim rather than stopping at the first plausible answer.

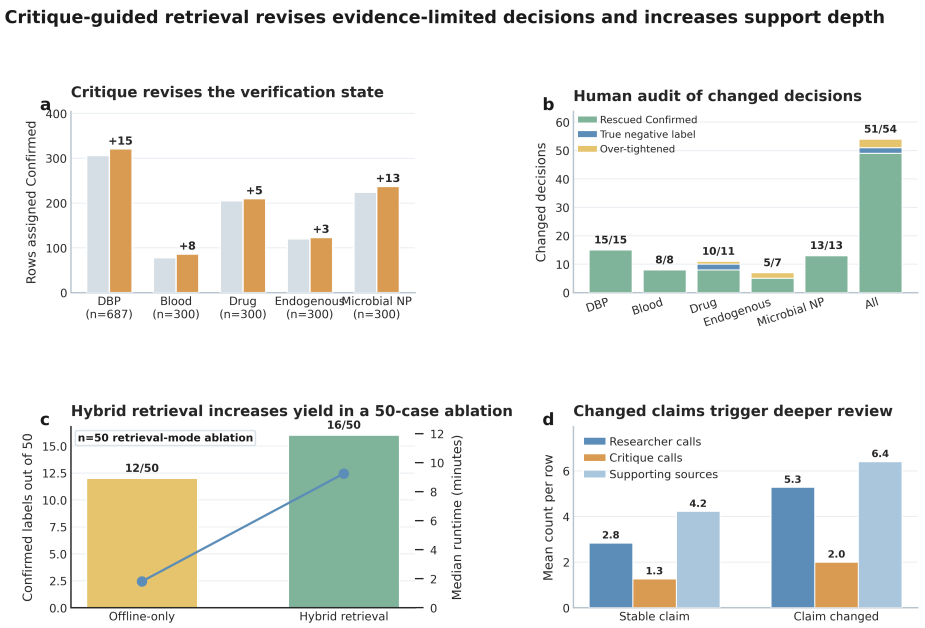


Figure 5 | Iterative retrieval and critique-agent utility.

A, Initial researcher claims versus final critique-loop decisions by topic; the left bar is the first researcher claim and the right bar is the final critique-loop decision. B, Human audit of confirmation-changing claim revisions, with topic-wise bars and a pooled All bar separating rescued Confirmed labels, true negative labels, and over-tightened outcomes. C, Offline-only versus hybrid retrieval in a 50-case mode ablation, jointly showing confirmation yield and median runtime. D, Mean exact researcher invocations, critique invocations, and supporting-source depth for rows whose claim stayed stable versus rows whose claim changed during the loop.

Operational Cost and Routing Behavior

We also quantified operational cost because database construction must scale beyond small demonstration sets. The cost profile shows a consistent pattern: straightforward Confirmed entries usually terminate after shorter evidence paths, whereas evidence-limited and suspected entries consume more tokens and time because the system continues searching until the relevant rubric gap is resolved or exhausted. Runtime closely follows token use across topics, making token budget a practical proxy for operational cost. Topic-dependent runtime variation reflects differences in literature accessibility and rubric specificity, while rubric-score distributions show that aggregate scores are useful diagnostics but do not replace predicate-level adjudication.

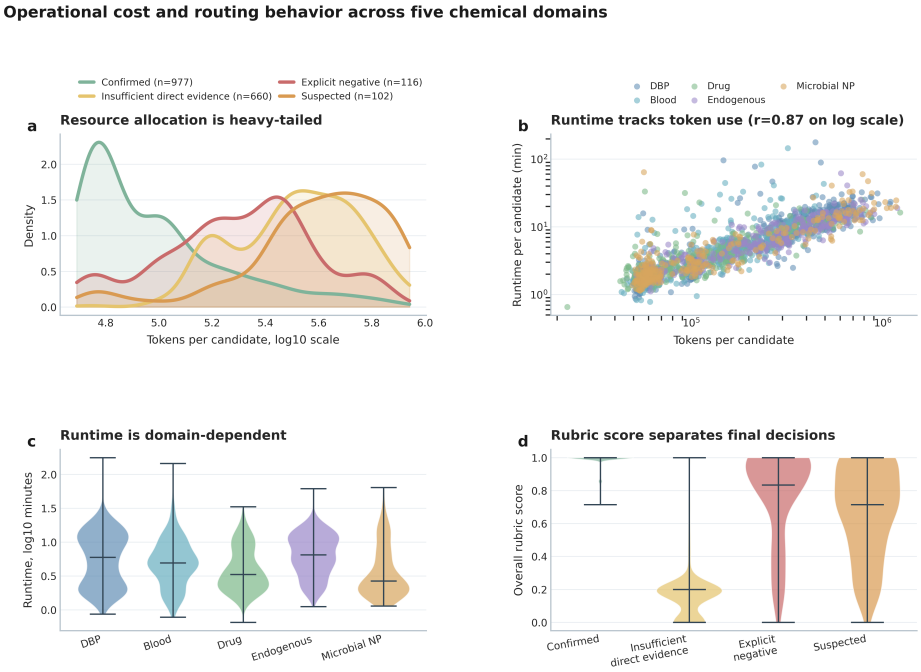


Figure 6 | Operational cost and routing behavior across five chemical domains.

A, Smoothed token-use density by final decision, showing heavy-tailed resource allocation. B, Per-row token use versus runtime by topic, showing the close coupling between compute and elapsed time on logarithmic axes. C, Runtime distributions by topic. D, Overall rubric-score distribution by final verdict. In this operational plot, "Insufficient direct evidence" denotes rows that remained below the Confirmed decision threshold.

Discussion

DB-Curator demonstrates that scientific database construction can be formulated as dynamic literature construction plus predicate-state verification. This framing differs from classical chemical information extraction, where the main objective is to detect and normalize entities from a fixed corpus.[13-18,34] It also differs from standard RAG, where retrieved context is usually used to generate an answer. Here, retrieved evidence changes the state of each candidate entry. A candidate can become Confirmed, remain evidence-limited, or be excluded when the retrieved context violates the user-defined rubric.

The closest methodological neighbors are scientific claim verification, agentic reasoning systems, and tool-using scientific assistants. Claim-verification benchmarks emphasize evidence retrieval and rationale selection.[28-31] ReAct, Reflexion, and deep-research agents show that LLMs can interleave reasoning, tool use, and iterative self-correction.[32,33,39,40] MRKL, Toolformer, and Voyager provide broader precedents for modular tool use and skill composition.[41-43] Chemistry-

focused systems have shown that LLMs and external tools can support molecular reasoning, extraction, and scientific discovery.[34-38] DB-Curator advances this line of work by organizing those capabilities around database-entry acceptance. The output is not a free-form answer or extracted mention. It is a structured row whose inclusion criteria, identity grounding, and supporting spans can be audited.

The fixed-cohort results show why this distinction matters. The zero-shot direct-generation baseline produced many acceptable compound names, but its citation trail rarely survived paper-level audit. This is a useful failure mode rather than a surprising weakness. It shows that familiar entity generation and release-quality curation are different tasks. DB-Curator recovered benchmark-overlap compounds, surfaced literature-supported compounds absent from the benchmark, and attached paper-level provenance to accepted rows. The benchmark-scale analyses then show that the same workflow applies across DBPs, blood exposome chemicals, drugs, endogenous metabolites, and microbial natural products. The domain-dependent Confirmed rates are expected because each topic has different evidence requirements and different literature coverage.

The rubric is central to this behavior. Instead of relying on an implicit model confidence score, DB-Curator converts the user specification into mandatory and exclusionary predicates. The Critique agent then audits claim–evidence alignment at the predicate level. This makes the stopping condition explicit: the loop stops when required evidence is found, when the candidate violates the rubric, or when the available evidence remains insufficient within the search budget. This structure reduces the risk that plausible but unsupported statements are promoted to database facts. It also makes human review easier, because reviewers inspect candidate identity, the rubric decision, and cited paper evidence rather than an unstructured generated answer.

The system still has limitations. First, DB-Curator prioritizes text and table evidence. Multimodal extraction from figures and supplementary materials would improve coverage, especially for quantitative measurements. Second, open-access limitations remain important. Some relevant papers cannot be fully inspected unless publisher access, author manuscripts, or user-provided PDFs are available. Future versions could integrate institutionally licensed access, where permitted, or local full-text collections supplied by users. Third, retrieval policy remains a source of performance variance. Improved adaptive query planning, selective follow-up search, tighter coupling between retrieval diagnostics and verification, and stronger penalties for reagents, solvents, and contextual chemicals could improve recall without relaxing acceptance standards. These policies are natural targets for reinforcement learning because downstream verification outcomes provide feedback on which retrieval actions improve evidence quality.

Although the evaluation focuses on chemistry and biomedicine, the architecture is intended to be transferable. Other domains would require new entity recognizers, grounding resources, and rubrics, but the control logic can remain the same. In materials science, predicates could cover composition, synthesis route, electro-

chemical setting, and reported performance. In clinical-trial curation, predicates could cover intervention, population, endpoint, study design, and statistical support. From this perspective, DB-Curator is a multi-skill agent workflow for scientific curation. Scope refinement, dynamic literature construction, entity grounding, candidate triage, predicate-level verification, and provenance generation are coordinated around the same acceptance contract. Overall, DB-Curator provides a pathway from on-demand literature retrieval to curated, machine-actionable databases with explicit scope specification, structured evaluation rubrics, multi-agent verification, and auditable provenance.

Methods

Overview

DB-Curator is a stateful multi-agent pipeline that maps a user request u into a curated topic database B. It is designed as a deep research agent in the sense that, rather than answering from a single context window or extracting from a fixed literature set, it iteratively searches, reads, and verifies evidence across a paper set that is built during the run. This design can be viewed as agentic retrieval-augmented verification: retrieval is part of the evolving reasoning state, and new searches are conditioned on the refined user definition, inspected search results, candidate identities, and unresolved rubric predicates. A run maintains a shared state containing the specification, planned queries, retrieved documents, retrieved chunks, candidate entities, evidence snippets, provenance pointers, and audit artifacts. Each node updates this state and writes intermediate outputs to disk so the run can be reproduced and inspected.

$$B = F(u).$$

The pipeline has four stages. Stage 1 performs human-in-the-loop intent refinement and converts the user request into a topic definition, inclusion criteria, exclusions, and verification rubric. Stage 2 performs broad literature harvesting, acquires full-text context when available, and discovers candidate chemicals through named-entity recognition and normalization. Stage 3 converts the noisy candidate set into a smaller verification queue by combining semantic topic similarity, compound grounding, and LLM candidate triage. Stage 4 verifies each candidate through a targeted Researcher–Critique loop that gathers evidence, checks every rubric item, and assigns a canonical claim label.

Definition of specialized agentic tools

DB-Curator executes through a small set of specialized tools with separate roles. These tools implement lower-level skills for literature discovery, chemical grounding, candidate extraction, evidence retrieval, and predicate-level verification. Together, they form the deep-research loop: broad discovery expands the evidence pool, NER extracts candidate chemicals, session indexing supports within-paper reading, and iterative querying changes the paper set and evidence set until a candidate is accepted, excluded, or left evidence-limited.

CompoundDatabaseTool. This tool is invoked only when the verification loop requires compound-specific grounding, for example when a candidate entity is ambiguous or when additional synonyms are needed to form robust follow-up queries. Its input is a raw chemical string. Its output is a canonical identity and a synonym set from PubChem, with optional properties such as formula and mass when available.

FullTextJournalSearchTool. This tool performs broad literature discovery over online sources and an offline corpus. It searches online sources and offline lexical metadata, then fuses source-specific rankings with reciprocal rank fusion. Its output is a set of retrieved papers with metadata and full-text, JATS, PDF, local-text, or abstract-derived content when available. Because it is called at multiple stages, the retrieved paper set is dynamic: broad topic searches support discovery, candidate-specific searches support entity verification, and critique-generated follow-up queries add papers targeted to unresolved predicates.

ChemicalNerTool. This tool performs chemical named-entity recognition and normalization during the upstream discovery phase. It is invoked after broad retrieval, when many documents have been collected and it would be inefficient to send all raw text directly to an LLM. The tool reads the full-text batch in parallel and distills it into a structured candidate list. It uses HunFlair2 for recognition and a linker module to map surface forms to canonical identifiers such as MeSH and PubChem CIDs. Its input is a directory of retrieved papers or cached text artifacts. Its output is a consolidated report listing identified chemicals, canonical identifiers, frequencies, and source document identifiers. This report supports objective selection and screening of candidates for downstream verification.

FullTextIndexTool. This tool builds a session-specific FAISS index over the full-text artifacts produced in the current run.[46] The output is an index that supports within-document search restricted to the retrieved set.

FullTextQueryTool. This tool performs deep reading for verification over the session index. Its output is a ranked list of evidence snippets with provenance. During verification it is used only for evidence seeking and does not introduce new candidates.

Specification and evaluation criteria refinement

We convert the user request into an operational specification s through a refinement agent. The specification defines inclusion criteria, exclusion criteria, and predicate definitions for what counts as an accepted entry. It is treated as a contract for all downstream agents.

$$s = f_{refine}(u)$$

The specification is then refined into a structured evaluation rubric that defines domain-specific criteria. These criteria can be reused across repeated runs on the same topic so that candidate-level verification is performed against a

stable contract. During verification, the Critique agent uses this rubric to make predicate-level acceptance decisions based on explicitly defined criteria for the domain.

In the current workflow, the human contribution is limited to clarifying scope and resolving ambiguous inclusion rules at the specification stage; subsequent retrieval, candidate generation, and verification are fully automated under the resulting rubric.

$$r = f_{rubric}(s, \text{domain})$$

For verification, the rubric is treated as a set of predicates rather than as a free-form instruction. For candidate e, each predicate $p_i \in r$ has an evidence state $z_{e,i} \in \{\text{supported}, \text{unresolved}, \text{exclusion-triggered}, \text{exhausted}\}$. Inclusion predicates must be supported, exclusion predicates must not be triggered, and unresolved mandatory predicates drive additional retrieval until the search budget is exhausted. This predicate-state representation is the control signal for the Researcher–Critique loop.

$$z_e = \{z_{e,i} \mid p_i \in r\}$$

A query generation component produces an initial set of broad search directions $Q_0(s)$ from the original query and refined definition. These directions are intentionally recall-oriented and are not converted into a hard domain-specific topic-keyword filter. The upstream agent first searches broad topic phrases, inspects the returned titles and abstracts, mines seed chemical classes and compounds, and then expands searches around those seed terms. This design avoids losing valid papers whose titles or abstracts omit the exact topic wording.

$$Q_0(s) = f_{harvest}(u, s)$$

During candidate verification, queries are generated differently. For a candidate entity e, DB-Curator builds compact structured searches from identity terms I_e, such as the canonical name and a small number of strong synonyms, and context terms $C_{e,r}$, which are derived from the current unresolved rubric item. This produces targeted query batches rather than repeated paraphrases of the same broad search.

$$Q_e = f_{verify}(I_e, C_{e,r})$$

Broad retrieval with hybrid online/offline fusion

Broad retrieval is executed by FullTextJournalSearchTool over both online sources and the offline literature corpus. Online sources include PubMed/PMC, Europe PMC, and the Semantic Scholar Academic Graph API,[7,51] with preprint sources

enabled only when the run configuration requests them. The offline corpus contains 6.3 million papers with local metadata and text. Reported full-loop runs used both online and offline retrieval, while the offline-only ablation disabled online APIs. The key distinction from fixed-corpus information extraction is that retrieval is invoked throughout the workflow. The upstream agent first builds a topic-level paper set for candidate discovery, and the downstream verifier later builds candidate-specific paper sets for evidence checking.

For each query q, the tool launches source-specific searches in parallel. Online searches return ranked paper records from their native APIs. Offline search uses SQLite full-text search over title, abstract, identifiers, and metadata.[45] These offline metadata hits are fused with online search results before full-text enrichment.

The retrieval unit at this stage is a paper record rather than a final evidence span. Records are deduplicated by stable paper identity, using DOI, PMID, PMCID, Semantic Scholar identifier, arXiv identifier, or normalized title. Source- and query-specific ranked lists are then combined by reciprocal rank fusion:

$$R(d) = \sum_{j \in J(d)} \frac{1}{k + rank_j(d)},$$

where $J(d)$ is the set of source-query runs that returned document d, $rank_j(d)$ is its rank in run j, and $k = 60$ in the default configuration. The reported full-loop settings kept retrieval broad and passed the fused set to full-text enrichment rather than relying on a narrow per-source cutoff.

$$D_{\text{retrieved}} = Keep_{\text{RRF}}\left(\{d\}, R(d)\right)$$

This dynamic retrieval design makes the literature collection query-conditioned and candidate-conditioned. Early broad searches maximize discovery recall, inspection-derived seed terms expand the candidate-generating corpus, and later critique-derived searches target the specific predicate that remains unsupported for a candidate.

Full text acquisition and parallel processing

For offline full-text acquisition, DB-Curator hydrates retrieved records from local text or PDF paths when those paths are available. For online sources, it first attempts structured JATS or PMC full text, then Unpaywall/open-access PDF routes, and finally PDF conversion. PDF text is extracted with OpenDataLoader[48] as the primary converter and Docling[47] or pdfminer as fallbacks when configured. When full text is unavailable, abstracts are retained as abstract-only context so that the paper can still contribute limited metadata-level evidence. The implementation preserves section headings when possible and supports parallel workers for source search, enrichment, PDF conversion, embedding, and reranking.

Candidate discovery by entity extraction, normalization, and filtering

Candidate discovery is performed after broad retrieval. ChemicalNerTool loads the retrieved paper JSON files in folder mode, extracts text from each file, and processes documents in batches. The NER backend uses HunFlair2 for chemical mention recognition and a biomedical linker for entity normalization. Mentions are associated with source files, and linked identifiers such as MeSH IDs are retained when available. This per-file representation is important because it keeps mention context available for later candidate ranking and audit.

We construct a raw candidate set by aggregating linked entities and surface forms across retrieved papers:

$$E = \bigcup_{d \in D_{\text{retrieved}}} \{norm(m) \mid m \in M(d)\}.$$

For each candidate e, DB-Curator stores mention counts, mention samples, linked identifiers, and surface forms. A representative candidate name is chosen from the observed surface forms and synonyms. When possible, CompoundDatabaseTool checks PubChem compound grounding, uses RDKit for structure handling, and uses OPSIN-based name-to-structure resolution when available.[5,49,50] These checks help prefer names that match concrete compounds over noisy mention strings, identifiers, or broad class labels.

Because the raw candidate pool contains off-topic chemicals, reagents, solvents, precursors, and contextual compounds, DB-Curator applies two filters before expensive verification. The first filter is deterministic pre-ranking. Mention snippets $S(e)$ are scored against the refined topic definition using the BAAI/bge-small-en-v1.5 embedding model, with a TF–IDF fallback when the embedding model is unavailable:

$$s_{\text{topic}}(e) = \max_{s_i \in S(e)} cosine(embed(s_i), embed(s)).$$

The candidate score also includes compound-grounding features from PubChem lookup and mention frequency. Candidates are sorted by topic similarity, grounding score, compound match score, and mention count. The second filter is an LLM candidate-triage step. A Gemini 2.5 Flash candidate-filtering prompt receives only the selected candidate names, synonyms, topic definition, and verification template. It accepts specific compound candidates that are plausibly in scope and assigns a relevance score. Candidates below the score threshold are not sent to downstream verification.

$$E' = \{e \in TopK(E) \mid LLM_{\text{filter}}(e, s) \geq \theta\}.$$

Dual-agent verification and provenance

Each candidate entity $e \in E'$ is verified in its own session. The verification payload contains the candidate name, synonyms, original topic request, refined definition, rubric items, and a compact search budget. The Researcher agent first grounds the entity with CompoundDatabaseTool when possible. This step expands safe synonyms and compound identifiers for search, but it does not add new candidates to the release queue.

Literature discovery during verification uses structured query batches. The query builder prioritizes one or two primary identity terms, then interleaves those identity terms with the current context terms. Additional aliases are kept discoverable but are not crossed with every context term, which prevents synonym explosion. In later rounds, the Critique agent supplies short follow-up queries tied to the highest-priority unresolved rubric item, and duplicate follow-up queries are suppressed.

After papers have been retrieved, FullTextIndexTool builds a session-specific FAISS index. Retrieved papers are split by section when section structure is available, otherwise by full text, using approximately 1200-character chunks with 180-character overlap. Trivial metadata stubs and very short non-substantive chunks are removed before embedding. Chunk embeddings are generated with all-MiniLM-L6-v2.

Evidence seeking is implemented with FullTextQueryTool over this session index. For each generated query, the tool retrieves a larger candidate pool using an ensemble of BM25 lexical retrieval and FAISS vector retrieval with equal weights when both are available. If BM25 is not available, vector retrieval is used alone. The default fetch size is the larger of 120 chunks or four times the requested output size. Candidate chunks are then reranked with BAAI/bge-reranker-v2-m3, with cross-encoder/ms-marco-MiniLM-L-6-v2 as a fallback. Scores are aggregated across query variants by keeping the best score for each chunk, and the top snippets are returned with title, section, DOI, PMID, and source-path metadata.

The Researcher agent synthesizes a claim from these snippets using the provided rubric. The Critique agent then audits whether the claim is supported by the cited evidence and whether every rubric item is solved. The workflow allows several researcher-only rounds before critique if the researcher has not yet produced a fully solved rubric assessment. After critique, one additional targeted researcher round may be executed when the Critique provides actionable follow-up queries. Otherwise, the loop stops when the Critique approves the claim, when the evidence supports rejection, when tools are unavailable, or when repeated follow-up does not add rubric-relevant evidence. To prevent unbounded agent debate, the graph is explicitly bounded. The reported full-loop evaluation runs used a downstream LangGraph recursion limit of 24 graph steps, and the Critique prompt requires one of three actions at each critique point: approve the claim, request a bounded targeted follow-up tied to a specific unresolved predicate, or emit a terminal evidence-limited or negative decision. Duplicate follow-up queries are suppressed, and repeated follow-up without rubric-relevant

evidence forces an exhausted terminal state.

$$y_e = v(e, T_e, r),$$

where T_e is the retrieved and indexed evidence state for entity e, r is the rubric, and y_e is a canonical claim label. The main labels used for analysis are Confirmed, Suspected, Explicit Negative Evidence, and Insufficient Direct Evidence. For accepted entities, DB-Curator stores the candidate identity, final label, rubric assessment, cited papers, and supporting snippets so that the final database can be audited at the evidence-span level.

$$B = \{(e, P_e) \mid e \in E', y_e = \text{Confirmed}\}$$

Evaluation framing

We use two complementary evaluation settings in this manuscript. The first is a fixed-cohort human-audit comparison over 50 DB-Curator confirmed rows and 50 zero-shot baseline rows per topic. In this setting, benchmark overlap is accepted automatically and non-overlap rows are resolved by manual inspection, so the comparison isolates whether each system produces releasable compound entries with auditable citations. The second is a benchmark-scale verification analysis over all available benchmark-positive rows for the five topics. In that setting, the primary outcome is whether each benchmark row reaches Confirmed under the topic-specific evidence rubric, with non-confirmed outcomes collapsed into an Insufficient Direct Evidence class for acceptance-rate analysis. These two settings answer different questions. The fixed cohort measures release readiness of generated entries, whereas the benchmark-scale analysis measures verification behavior over supplied benchmark compounds.

The baselines and ablations are interpreted according to the component they isolate. Because open-ended database construction does not have a standardized retrieve-then-generate protocol, and because retrieval choices would themselves define a new curation system, we use component-isolation analyses rather than an arbitrary retrieve-then-generate baseline. The zero-shot direct-generation baseline tests direct generation from the topic definition without retrieval, compound grounding, or critique. The zero-shot benchmark-labeling baseline tests whether an LLM can apply the same topic definition to supplied benchmark compounds and provide citations. The first researcher claim before critique is used as a researcher-only pre-critique state. The final Researcher–Critique label tests the full predicate-state loop with critique-triggered follow-up retrieval. The offline-only versus hybrid comparison isolates retrieval source breadth while keeping the downstream verification procedure fixed. Together, these comparisons separate direct generation, retrieval breadth, and predicate-level critique rather than treating all improvements as a single retrieval effect.

Citation audit and manual inspection

Citation audits were performed at the cited-paper level. A cited item was first resolved to a primary-paper record using DOI, PMID, PMCID, title, source-path metadata, or publisher, PMC, and PDF routes when available. For each resolved paper, we checked whether the quoted evidence span was present in the recovered text. When exact matching failed because of PDF conversion, formatting, or section-boundary differences, we accepted local-context support only if the same candidate and the same rubric-relevant support meaning were present in the recovered surrounding text. Compound-database identity pages and lookup records were excluded from primary-paper citation counts.

Manual inspections were performed by C.Z. and B.L. using the user-defined topic definitions, rubric criteria, and released audit sheets. Benchmark-overlap rows were accepted automatically, and non-overlap rows were reviewed for topic alignment, false compounds, out-of-scope contexts, and cases where literature supported a row despite a negative LLM comment. These inspections were not blinded to system identity; we therefore release the audit sheets, decision rules, and intermediate overlap and citation files to support external re-review.

LLM settings and reproducibility controls

All LLM calls used fixed prompt templates stored with the experiment code. Unless otherwise stated, generation used deterministic or low-temperature decoding. The evaluated Researcher model, Gemini 2.5 Flash, the Critique model, GPT-5-mini, and the zero-shot benchmark-labeling and citation-support baselines used temperature 0.0. Search budgets, RRF settings, stopping criteria, candidate-filter thresholds, and per-topic definitions were fixed within each reported evaluation run. The full benchmark-scale runs used a downstream recursion limit of 24 graph steps, two consecutive no-progress critique rounds before exhaustion, up to four follow-up query variants per round, a per-query literature search cap of eight records, up to six mechanical full-text search query variants, RRF filtering that retained 30–50 papers, and up to eight passage-query variants for indexed full-text querying. Random subsets for the 300-case benchmark domains and the 50-case retrieval-mode ablation were generated with fixed seeds and released with the evaluation artifacts.

Data availability

The offline literature corpus used in this study is derived from PubMed and Europe PMC and contains 6.3 million full-text articles. The processed offline dataset, including document identifiers and text chunks used for retrieval, is released publicly on Zenodo. Benchmark evaluation results, human audit labels, paper-audit summaries, and overlap analyses are released with the manuscript repository.

Code availability

The source code for DB-Curator is released publicly on GitHub under CZhao97/db_form_agent_system. The repository includes the multi-agent orchestration pipeline implemented with LangGraph, hybrid retrieval

components, chemical NER tools with HunFlair2 integration, verification logic for the dual-agent loop, and comprehensive provenance logging. Key files include `code/agent_graph_v2.py` for graph orchestration, `code/prompt_payloads.py` for structured rubric generation and evaluation criteria creation, `code/candidate_ranking.py` for semantic relevance filtering, `code/searching_tool.py` for hybrid retrieval over online APIs and the offline corpus, and `manuscript/evaluation_plan/` for the human-labeled evaluation, overlap, citation-audit, and figure-generation scripts. The repository also includes reproducible experiment scripts, configuration files for all five evaluation topics (DBP, blood exposome, drug compounds, microbial natural products, and human endogenous metabolites), instructions for downloading the Zenodo dataset, and complete environment specifications including dependency versions and model identifiers (Python 3.9+, LangGraph 0.x, sentence-transformers 2.x, and HunFlair2).

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Author contributions

C.Z., B.L., X.L., and T.H. conceived the study. C.Z. developed the software system and conducted experiments. C.Z. and B.L. performed the manual result inspection and evaluation analyses. C.Z., B.L., X.L., and T.H. wrote and revised the manuscript. X.L. and T.H. supervised the project.

Competing interests

The authors declare no competing interests.

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